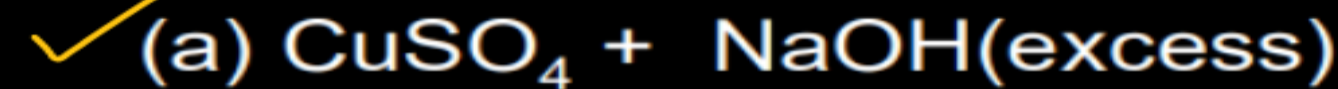


Ans: (3)

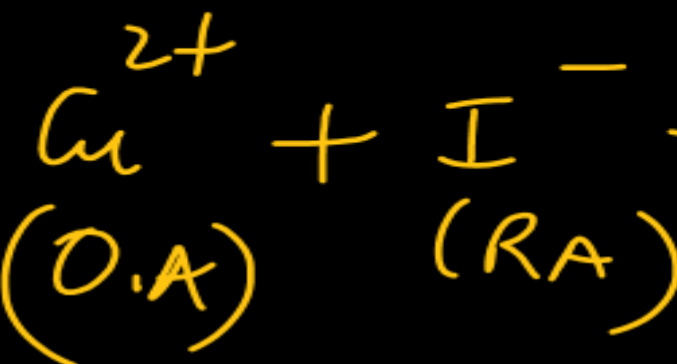
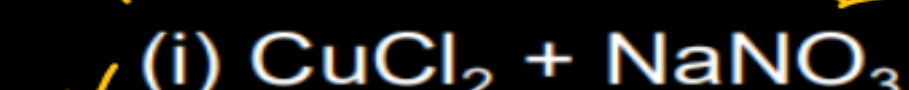
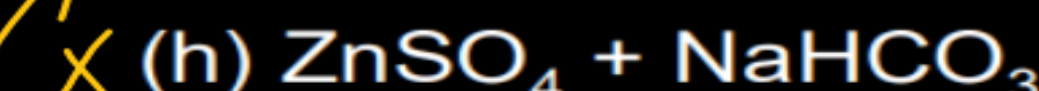
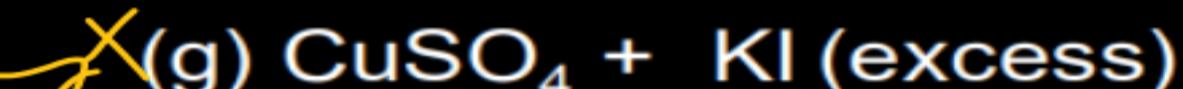
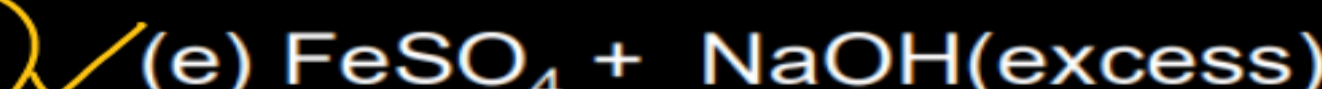
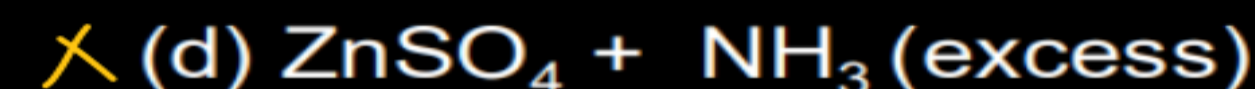
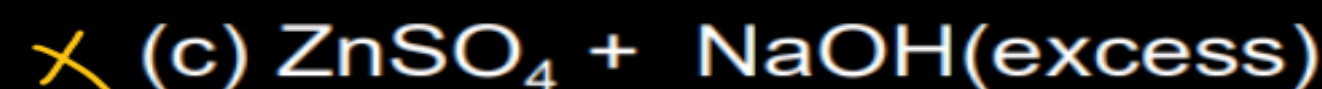
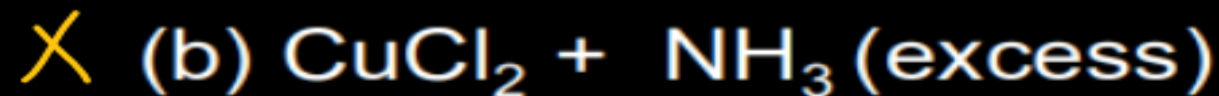
In how many of the following reactions, colored precipitate is obtained?



Blue

excess

no dissolution



white

+ $\text{I}_3^- (\text{aq})$



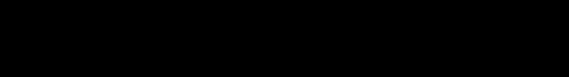
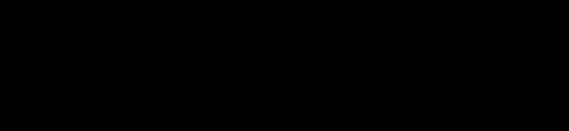
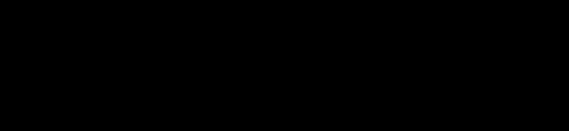
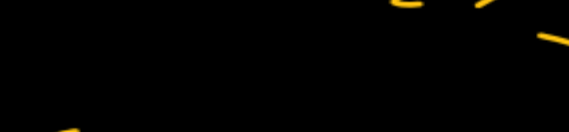
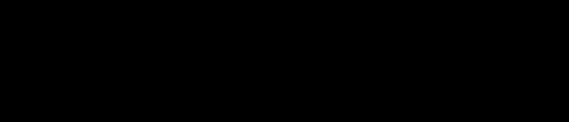
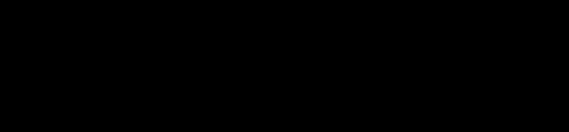
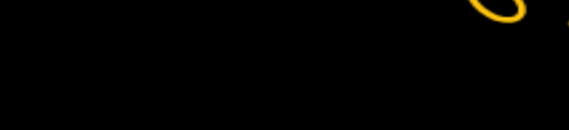
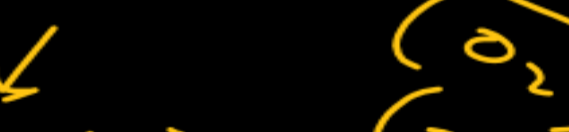
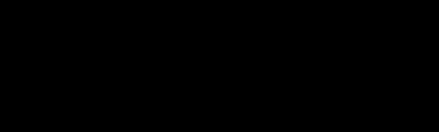
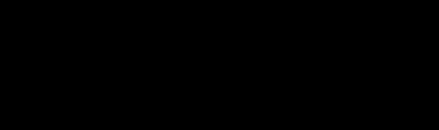
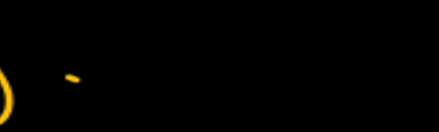
Reddish brown

excess

no dissolution



Reddish brown



of NaOH

Blue

excess

of NH_3

excess

OH^-

$\text{NH}_3(\text{excess})$

air

O_2

O.A

OH^-

excess

no dissolution

no dissolution

no dissolution

excess

of NH_3

excess

OH^-

$\text{NH}_3(\text{excess})$

air

O_2

O.A

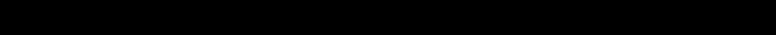
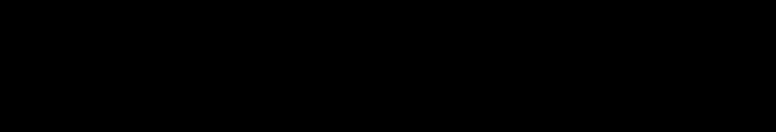
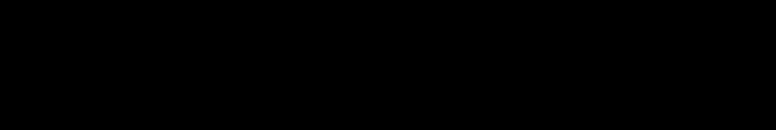
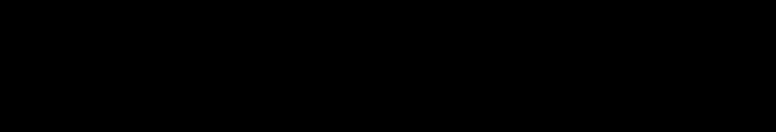
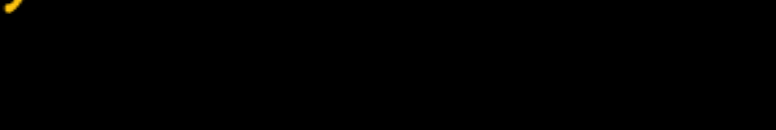
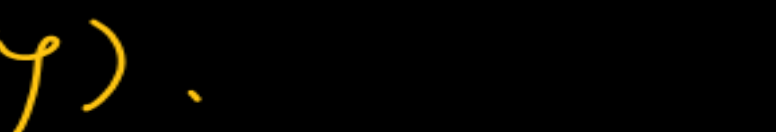
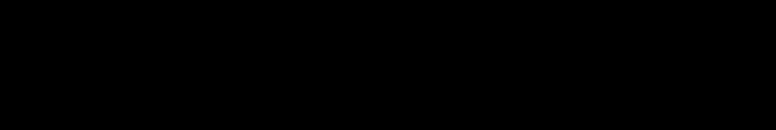
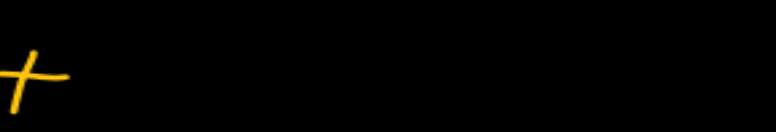
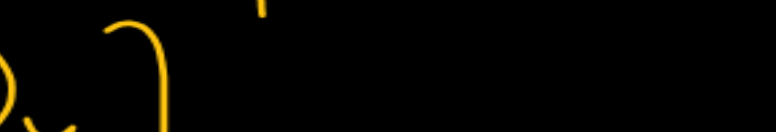
OH^-

excess

no dissolution

no dissolution

no dissolution



dsp^2 , sq. planar.

deep blue solution.

ZnO_2^{2-}

ZnO_2^{2-}

ZnO_2^{2-}

ZnO_2^{2-}

ZnO_2^{2-}

ZnO_2^{2-}

ZnO_2^{2-}

ZnO_2^{2-}

ZnO_2^{2-}

excess

of NH_3

excess

OH^-

$\text{NH}_3(\text{excess})$

air

O_2

O.A

OH^-

excess

no dissolution

no dissolution

no dissolution

no dissolution

no dissolution

excess

of NH_3

excess

OH^-

$\text{NH}_3(\text{excess})$

air

O_2

O.A

OH^-

excess

no dissolution

no dissolution

no dissolution

no dissolution

no dissolution

excess

of NH_3

excess

OH^-

$\text{NH}_3(\text{excess})$

air

O_2

O.A

OH^-

excess

no dissolution

no dissolution

no dissolution

no dissolution

no dissolution

excess

of NH_3

excess

OH^-

$\text{NH}_3(\text{excess})$

air

O_2

O.A

OH^-

excess

no dissolution

no dissolution

no dissolution

no dissolution

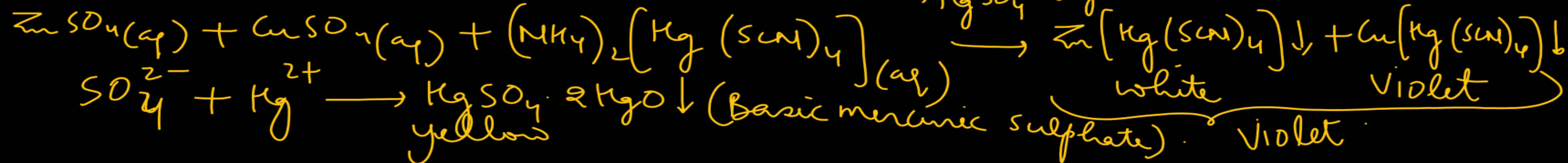
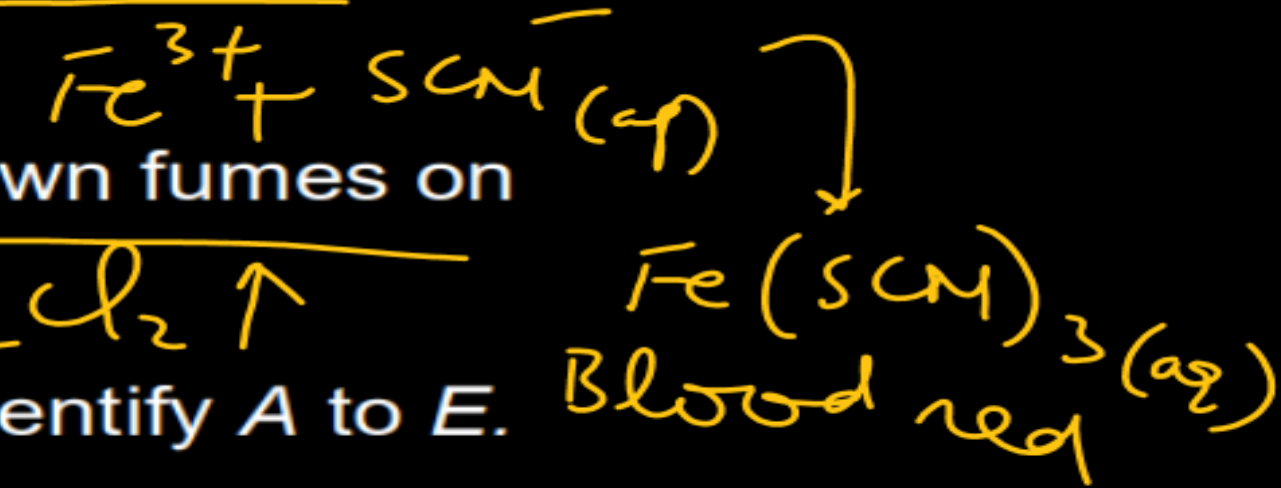
no dissolution



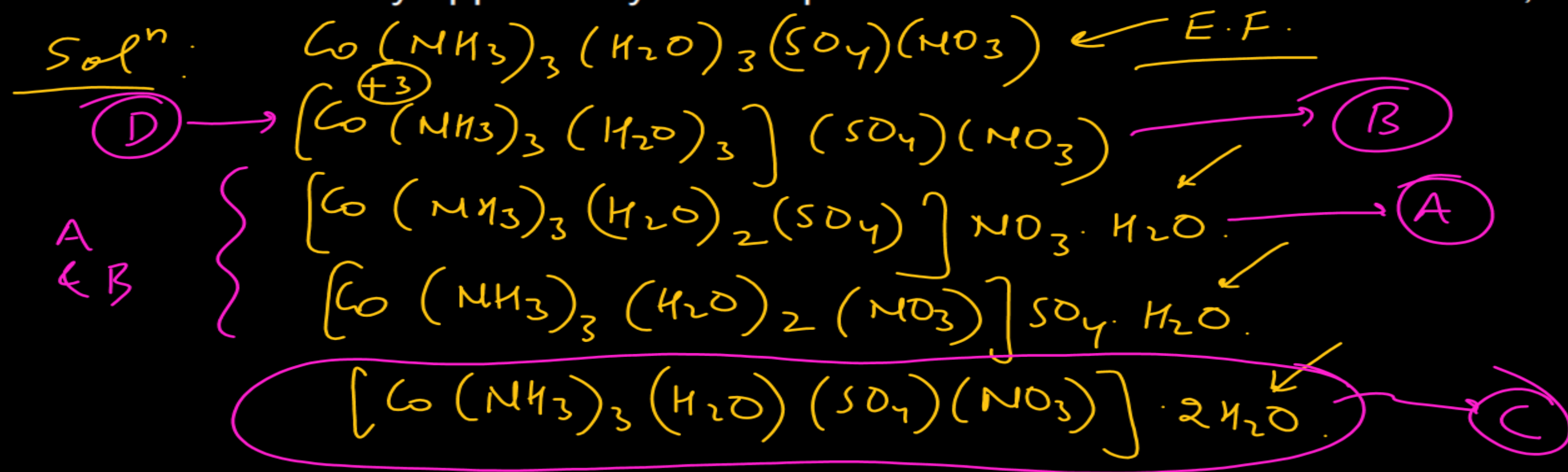
A mixture consists of an yellow salt (A) and a white salt (B), both anhydrous. Salt mixture was dissolved in water and few drops of HCl was added to obtain a clear, yellow coloured solution. The solution was then treated with $\text{NH}_3/\text{NH}_4\text{Cl}$ solution when a reddish-brown precipitate (C), insoluble in NaOH solution, was produced. Solution was filtered off and filtrate was treated with aqueous NaOH solution when a white precipitate (D), soluble in excess reagent, was formed. D was dissolved in dilute H_2SO_4 solution and few drops of copper (II) sulphate was added to it. To the resulting solution, 2.0 mL of ammonium tetrathiocyanato mercurate (II) solution was added when a violet precipitate was formed. Also precipitate (C), dissolved in dilute HCl and became blood-red coloured when NH_4SCN solution was added. Sodium extract of the original salt mixture gave the following results:

(a) Extract solution in concentrated H_2SO_4 , when mixed with $\text{K}_2\text{Cr}_2\text{O}_7$, gave a dense brown fumes on boiling, that turned aqueous NaOH solution yellow.

(b) Extract solution when treated with $\text{Hg}(\text{NO}_3)_2$, a yellow precipitate (E) was formed. Identify A to E.

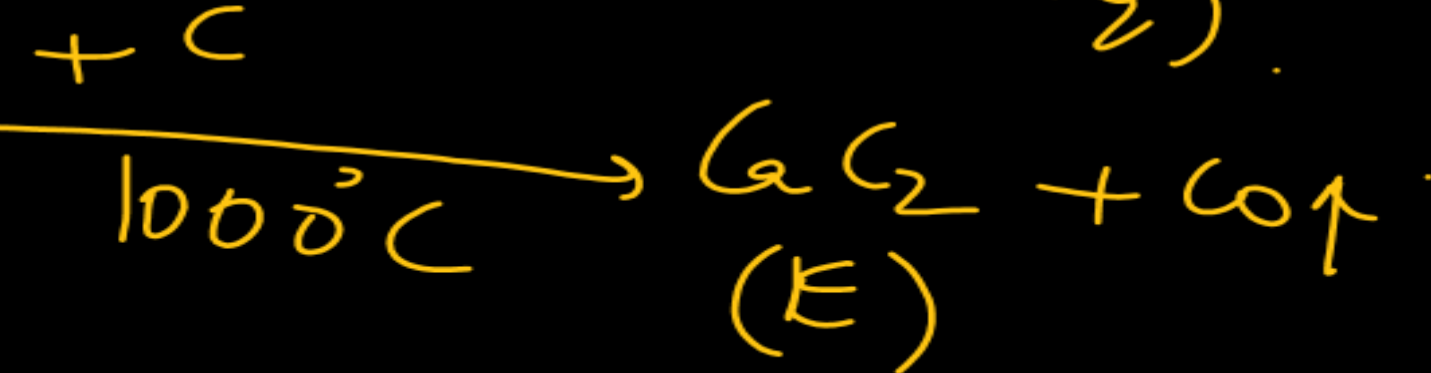
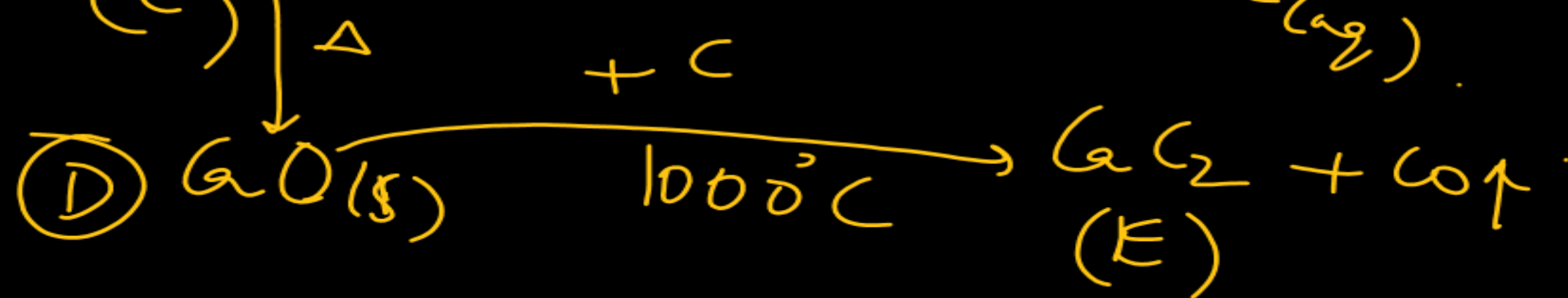
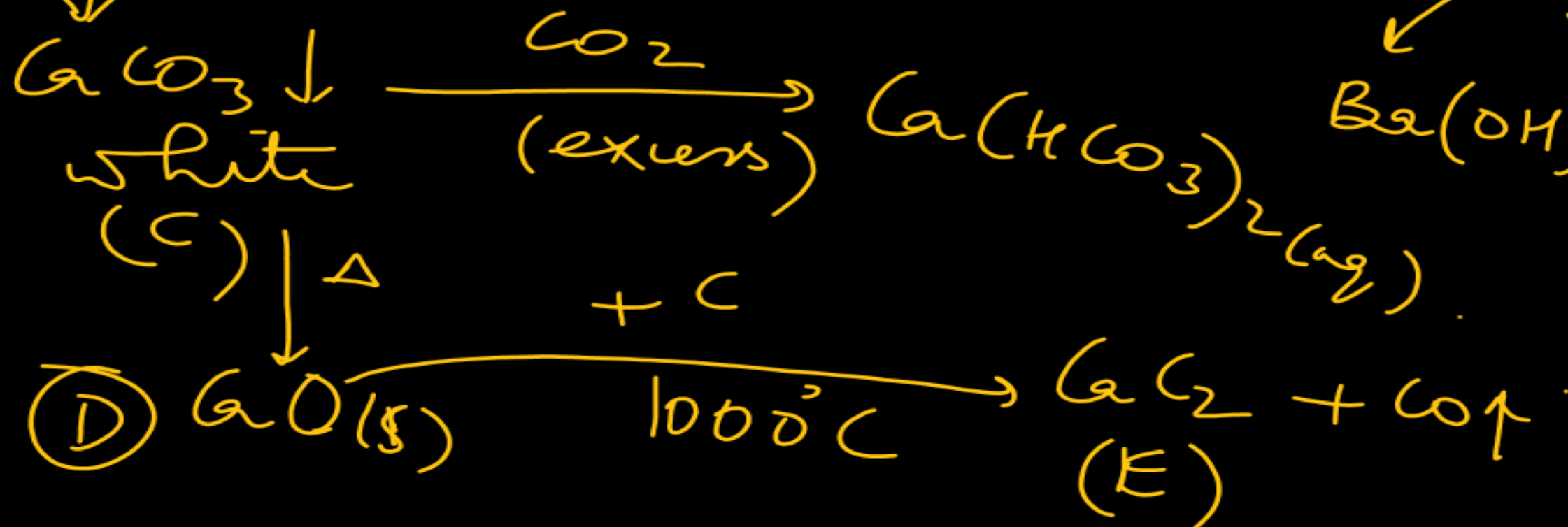
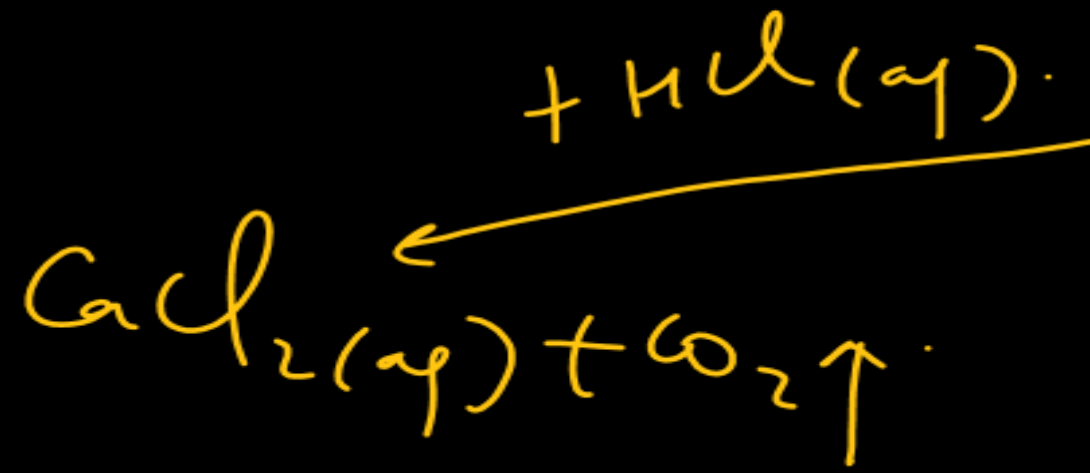
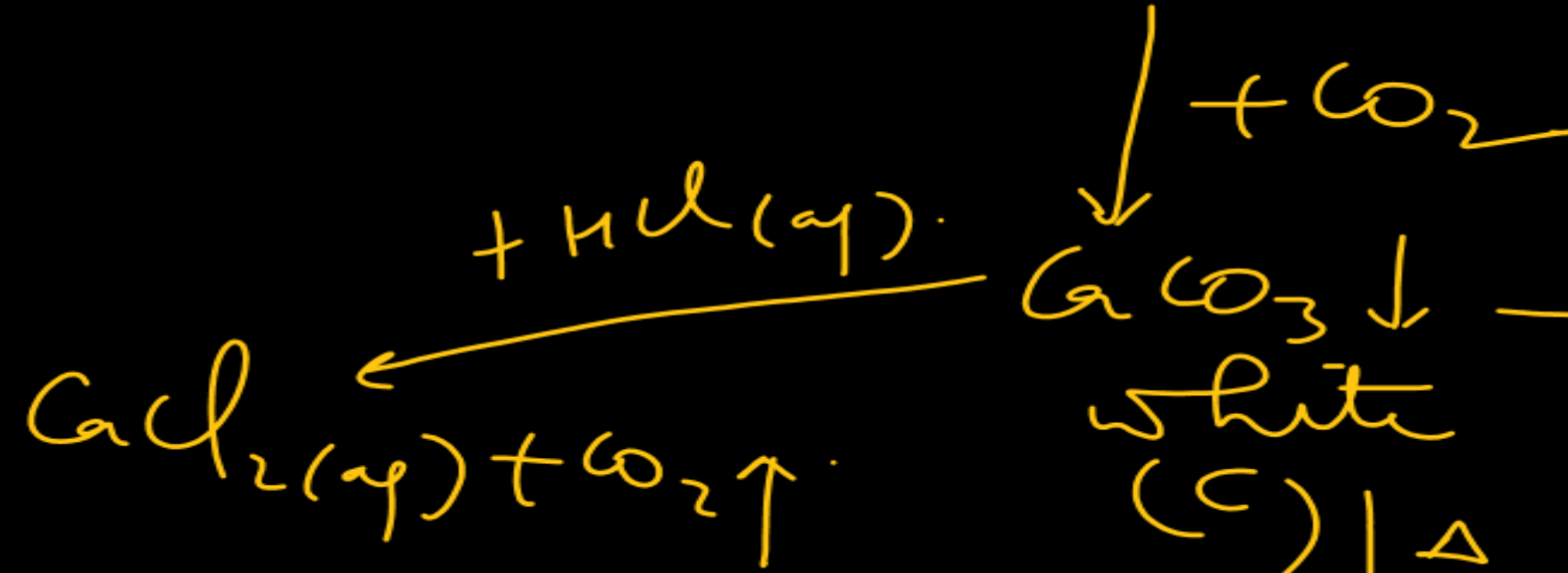
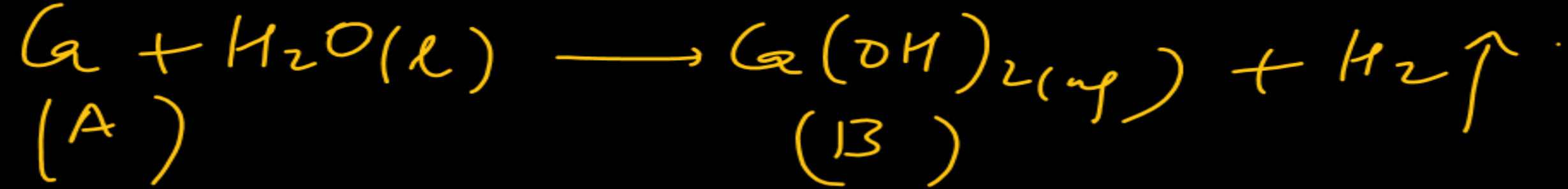


An octahedral complex of cobalt has its composition $\text{Co}(\text{NH}_3)_3 (\text{H}_2\text{O})_3 (\text{SO}_4) (\text{NO}_3)$ and it exists in four isomeric forms A, B, C and D. Simple heating of either A or B loses 5.6% of its original weight whereas similar treatment on C causes 11.2% loss in its original weight. D did not lose any weight on simple heating. Either A or C did not result in any precipitation if treated with aqueous BaCl_2 solution, whereas B and D resulted a white precipitate when treated with aqueous BaCl_2 solution. Also C does not conduct electricity appreciably in its aqueous solution. Deduce formula of A, B, C and D.



An element (A) ^{Ca} reacts quietly with water liberating a colourless, odourless gas ^{H₂} and a solution of (B).
 Passing CO₂ gas through B yields a white precipitate (C) which dissolves on passing excess CO₂ gas. ^{Ca(OH)₂ (aq)}
Precipitate C effervesced with dilute HCl and gave a brick red colouration to a Bunsen burner flame.
 Heating C gave a caustic white compound (D) which when further heated with carbon at 1000°C, gave a solid (E) of immense commercial importance. Identify A to E.

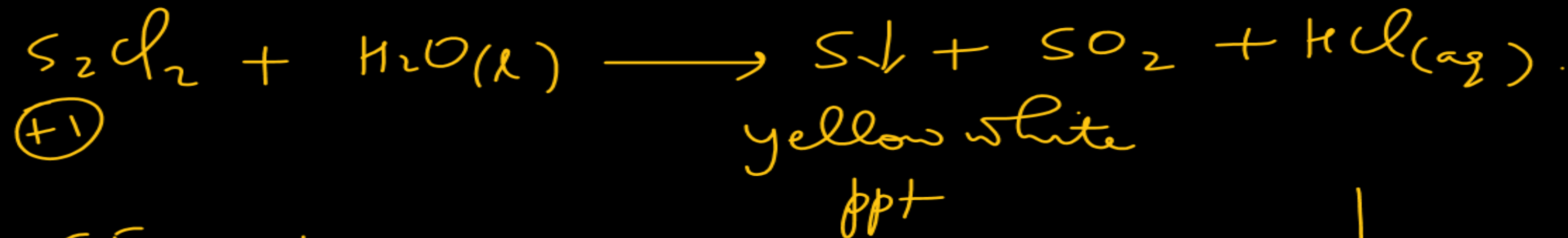
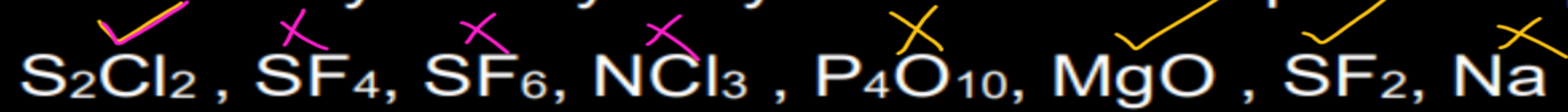
Solⁿ:



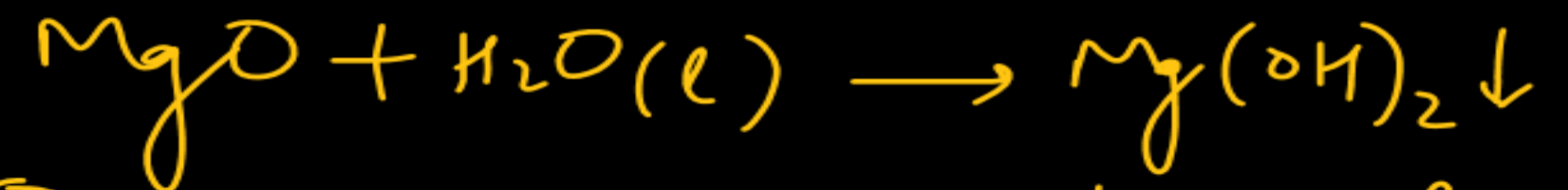
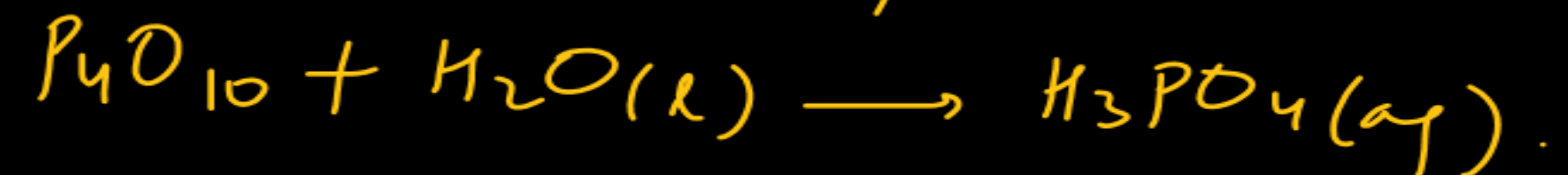
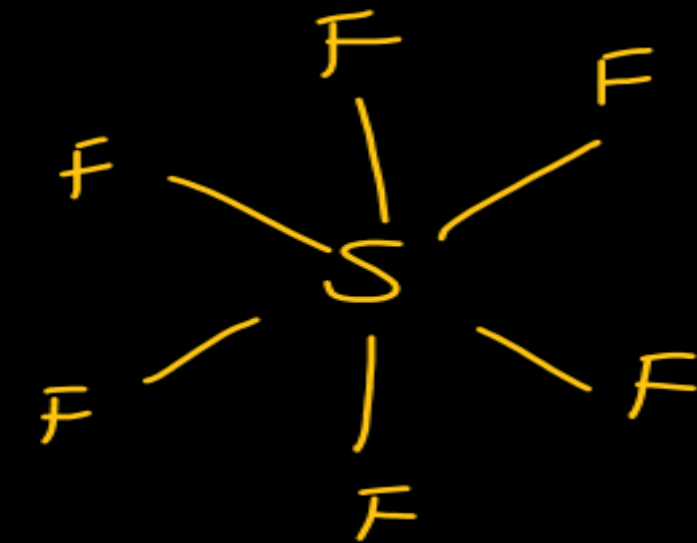
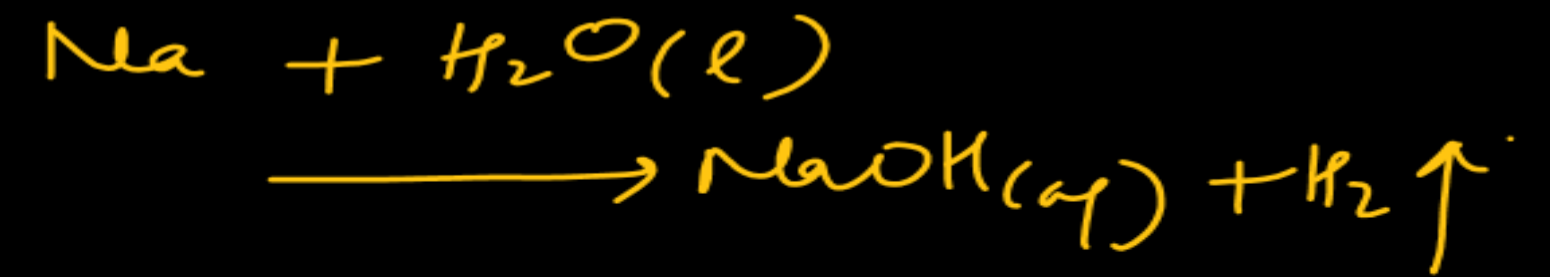
Lime water test
 $\text{Ca(OH)}_2\text{(aq)}$

Baryta water
 $\text{Ba(OH)}_2\text{(aq)}$

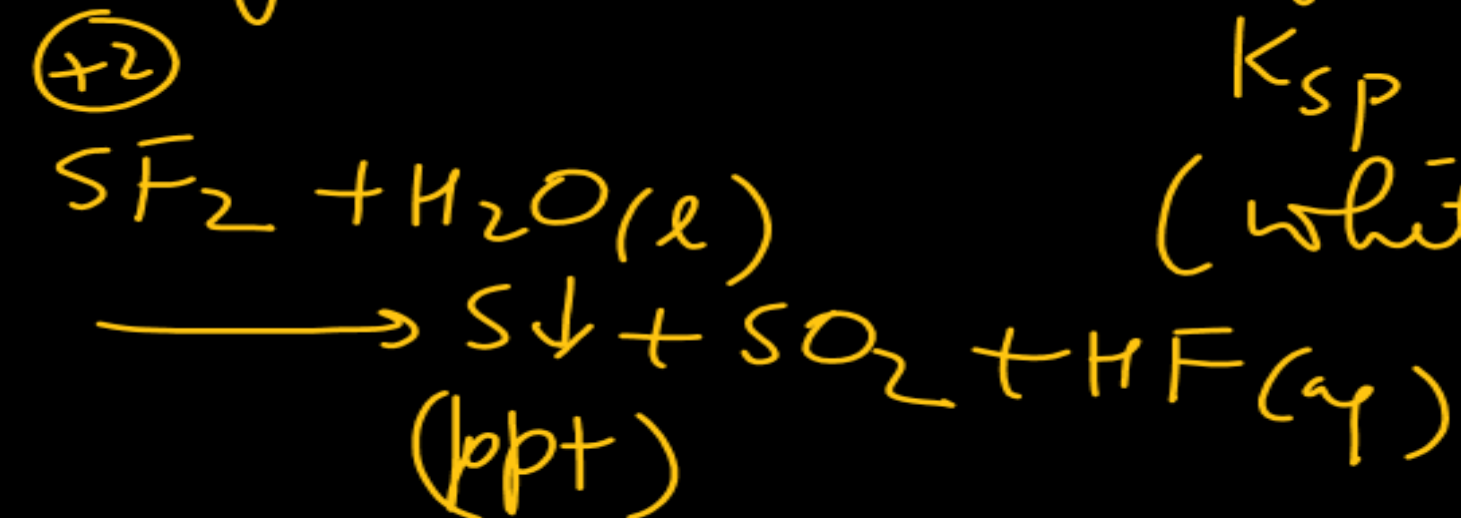
How many on hydrolysis with water produce precipitate? Ans: (3)

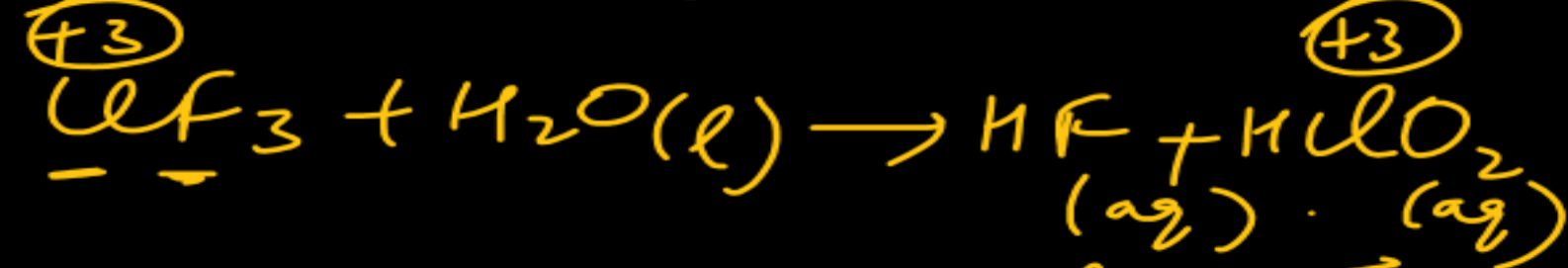
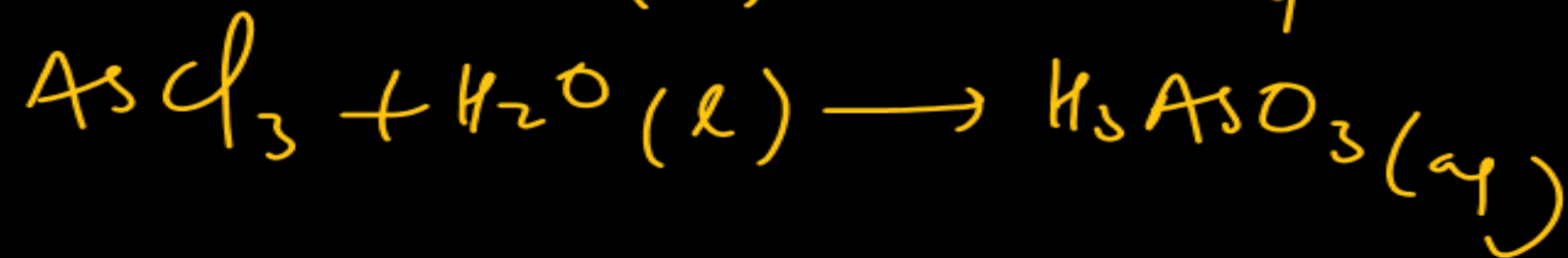
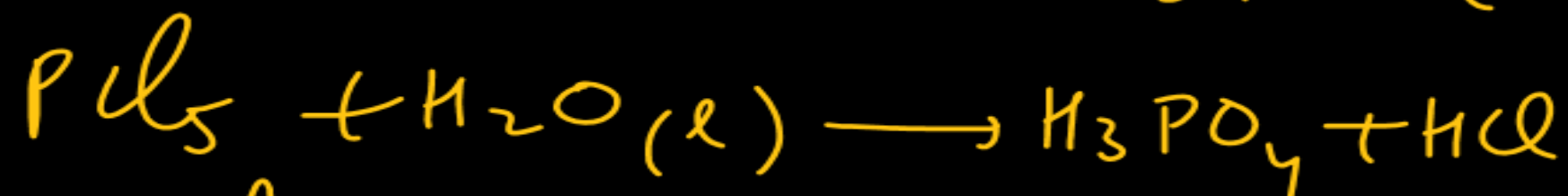


$\uparrow$
 an inert compound

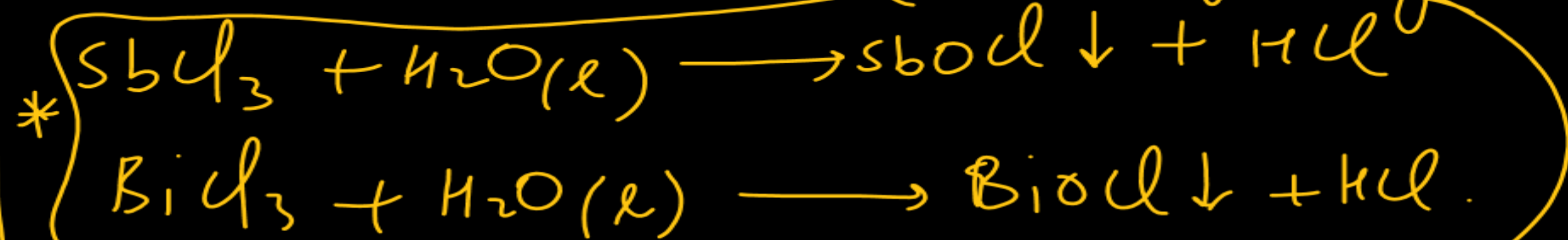


K_{sp} is low
 (white ppt)





No complete hydrolysis.



First species contains at least one weaker covalent bond than that present in second species in which of the following

(A) Diamond, graphite

(B)  (propane),  (cyclopropane) (***)

(C) SO₃, SO₄²⁻ X

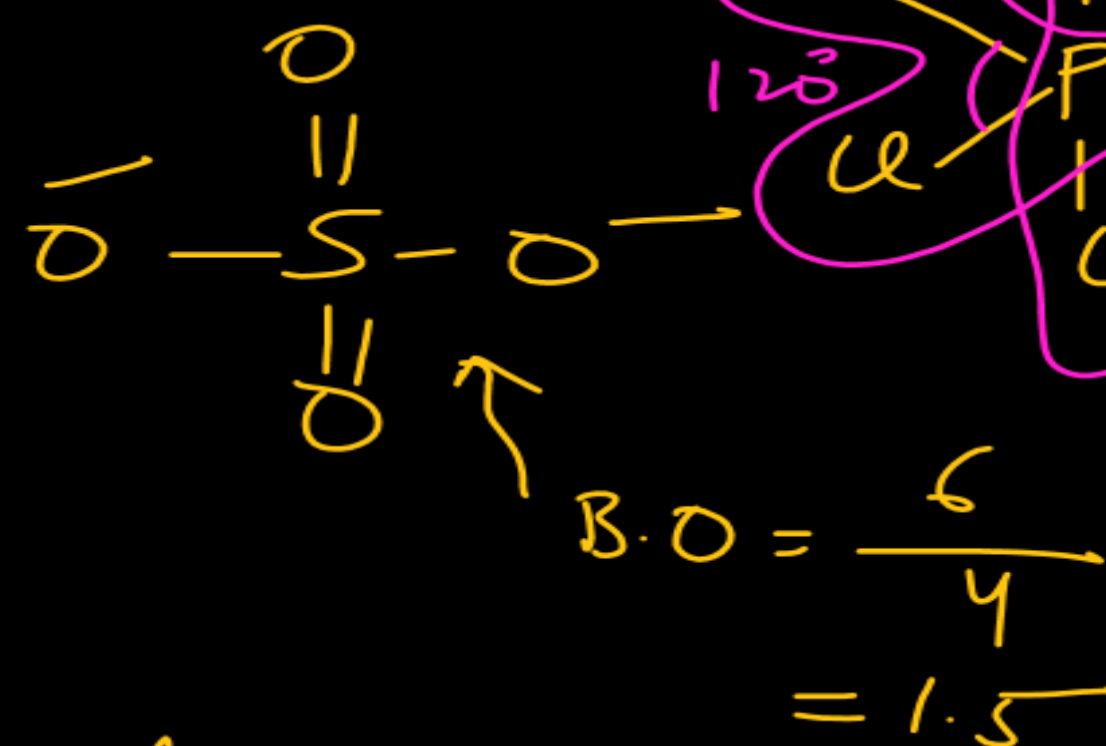
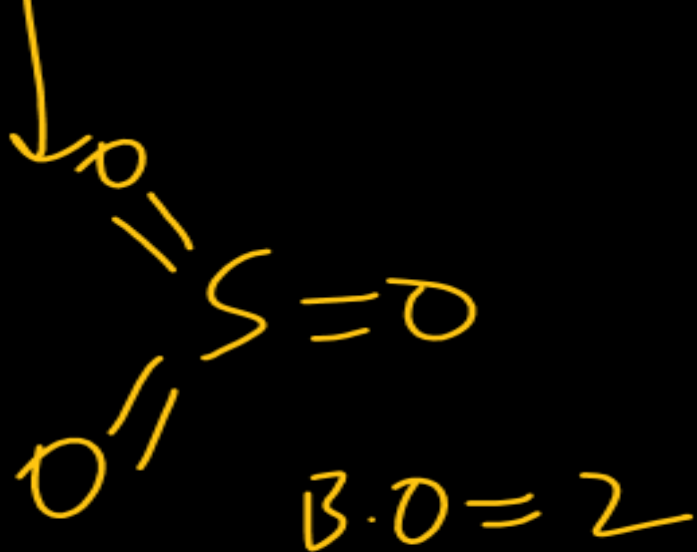
(D) PCl₅, PCl₆⁻

(C - C bond)

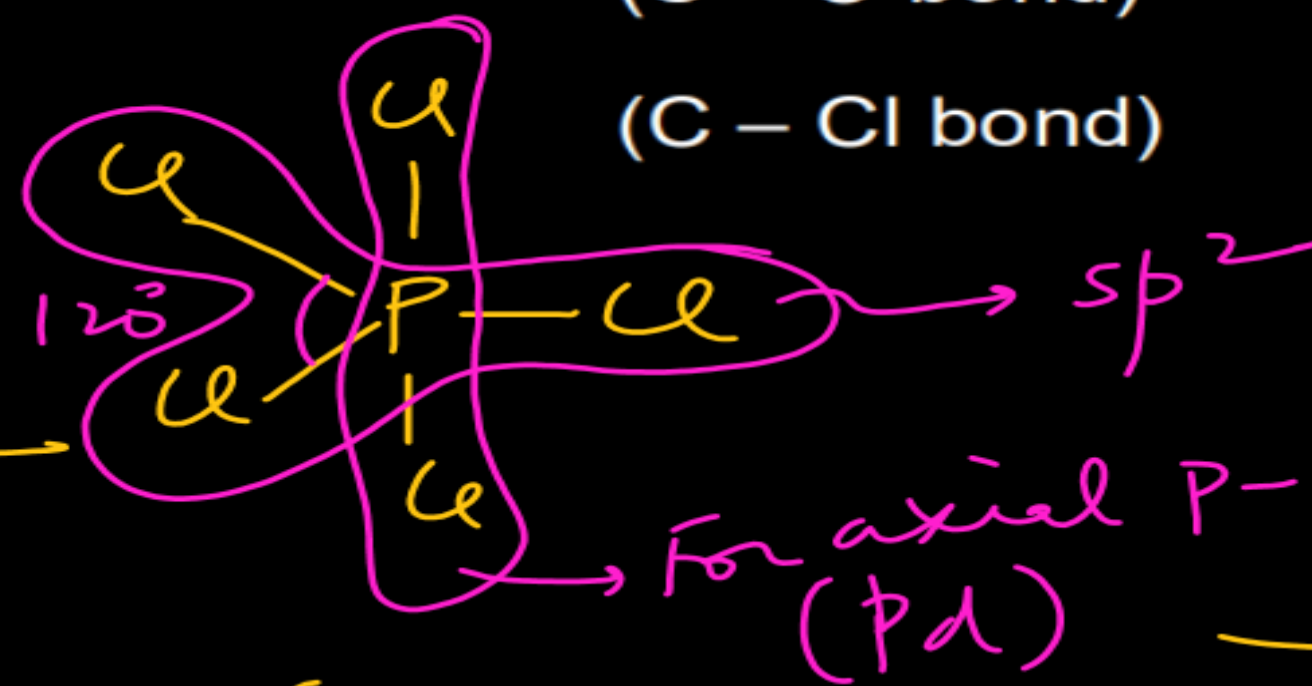
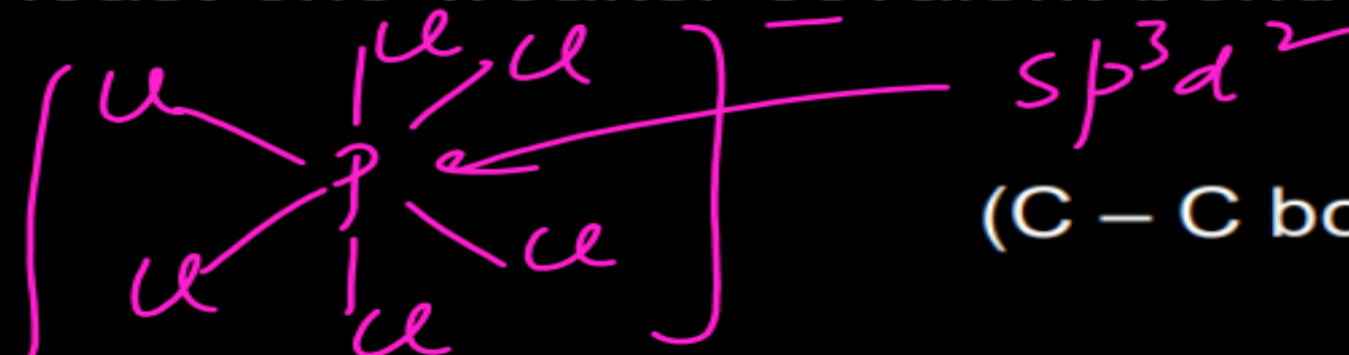
(C - H bond)

(S - O bond)

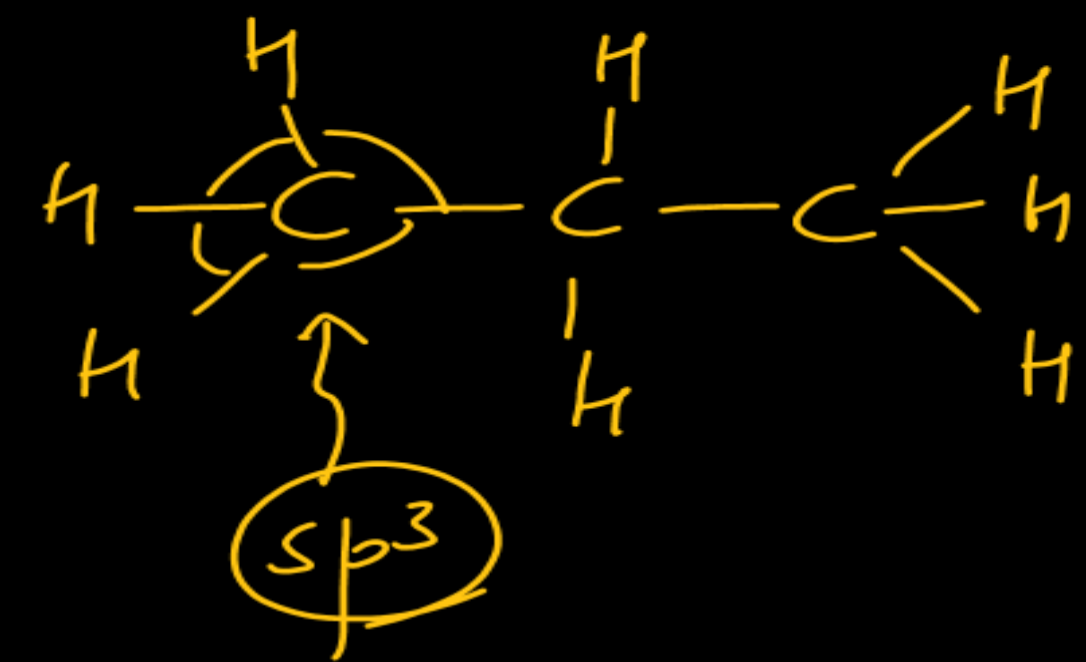
(C - Cl bond)



Ans: A, B, D



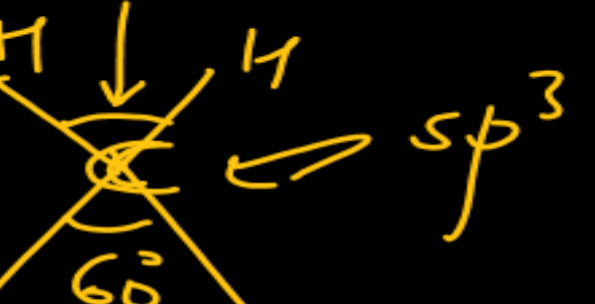
V-Good



Bent's rule

s ↑, B.A ↑

θ > 109°28'

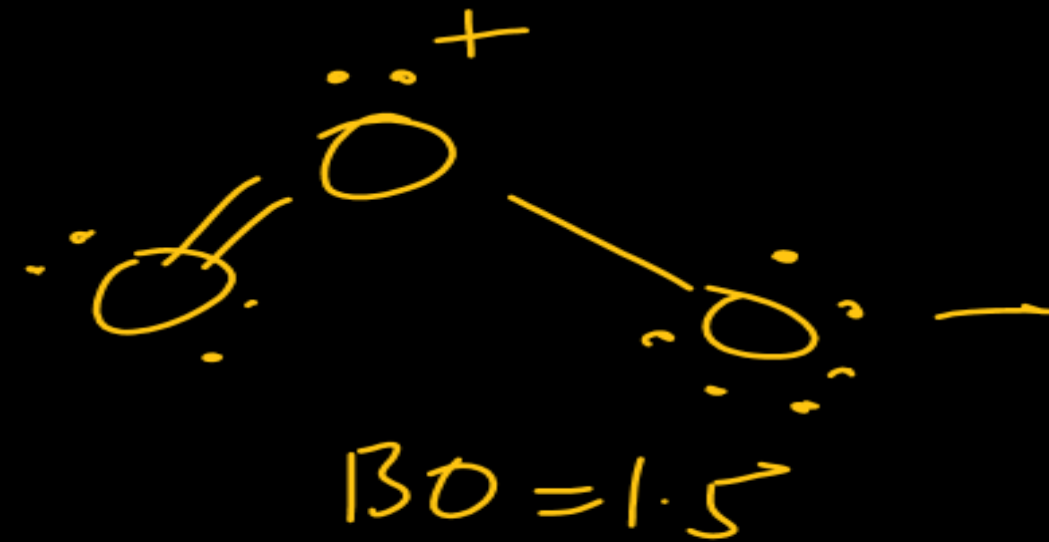


→ % s character in bond ↑, strength of bond ↑.
→ C-H bond of cyclopropane is stronger than that in propane.

Select the correct order(s).
 (A) $\text{PH}_3 < \text{AsH}_3 < \text{NH}_3 < \text{SbH}_3$: bond angle
 (C) $\text{O}_2 < \text{O}_3 < \text{O}_2^{2-}$: O-O bond length

$\text{BO} = 2$ $\text{BO} = 1.5$ $\text{BO} = 1$

$\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$



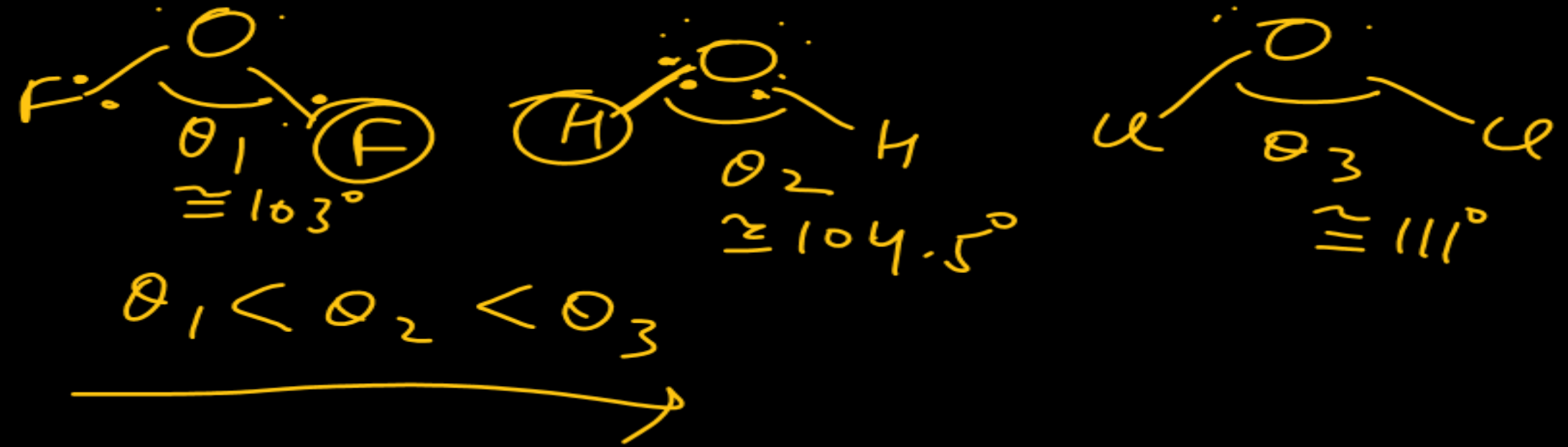
$$BL \propto \frac{1}{\text{BO}}$$

Ans: B, C

(B) $\text{F}_2\text{O} < \text{H}_2\text{O} < \text{Cl}_2\text{O}$: bond angle

(D) $\text{BF}_3 > \text{BCl}_3 > \text{BBr}_3$: bond angle

all $\text{BA} = 120^\circ$ ideal geometry



Match the species given in column – I with the characteristic(s) given in column -II

Column-I

(A) Orbital(s) used in sp^3d hybridization include

(B) Hybridization involving pentagonal bipyramidal geometry employs

(C) sp^3d^2 hybridization employs

(D) Orbital used by C atom for π – bonding in CO_3^{2-} (ion in x-y plane)

Column-II

(p) d_{xy}

(q) d_{yz}

(r) $d_{x^2-y^2}$

(s) d_{z^2}

(t) p_z

$sp^2 \rightarrow ns + np_x + np_y$ $\xrightarrow{z}$ internuclear axis

$sp^3 \rightarrow ns + np_x + np_y + np_z$

$sp^3d \rightarrow ns + np_x + np_y + np_z + nd_{z^2}$

$sp^3d^2 \rightarrow ns + np_x + np_y + np_z + nd_{z^2} + nd_{x^2-y^2}$

$sp^3d^3 \rightarrow ns + np_x + np_y + np_z + nd_{z^2} + nd_{x^2-y^2} + nd_{xy}$

(t)

